Biomaterials

生物材料

南方科技大学材料科学与工程系2024年春季课程

Testing Biomaterials

A: Introduction

B: In vitro assessment of tissue compatibility

C: In vivo assessment of tissue compatibility

D: Testing of blood-materials interactions

Testing Biomaterials

- How to characterize the materials that will be processed into a medical device/implant?

- How biomaterials can be evaluated to determine if they are biocompatible?

- How biomaterials can be evaluated to determine whether they function appropriately in the *in vivo* environment?

- How can testing criteria be defined to properly evaluate a given biomaterial's application?

  ---Some biomaterials complete their intended function in seconds

  ---Others are implanted for lifetime (10-70 years?)

Testing Biomaterials

- Testing always leads to experimental variability, particularly tests in living systems.

-The more complex the system (e.g. humans vs. cultured cells) the larger the variability that might be expected.

- Statistics should be used at two steps in testing biomaterials.

- Before an experiment is performed, statistical experimental design will indicate the minimum number of samples that must be evaluated to yield meaningful results.

- After the experiment statistics will help to extract maximum useful information.

- Detailed protocols are provided by:

- ASTM (American Society for Testing and Materials) and the
- ISO (International Standards Organization)
- FDA (Food and Drug Administration)
- NIH (National Institute of Health)
- The EU has its own directives in addition to the ISO standards.

Sometimes individual EU-member states has additional, national demands.

Category of Devices

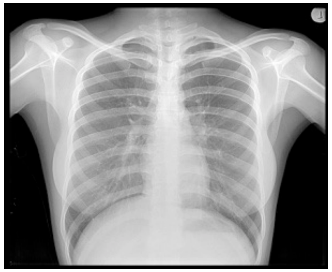
Biocompatibility tests depend on the category of devices:

- I. Non-contact devices
- II. External contact devices

- *Intact surfaces*: contact with external body surfaces only e.g., electrodes, external prostheses, monitors

- *Breached or compromised surfaces* e.g., burn dressings, wound-healing devices, occlusive patches

- III. Externally communicating devices



- *Intact natural channels* e.g., contact lenses, urinary catheters, colonoscopes

- *Blood path, indirect*: conduit for fluid entry to vasculature e.g., solution and blood administration sets

- *Blood path, direct:* contact circulating blood e.g., IV catheters, dialyzers and tubing, oxygenators and tubing

- IV. Internal devices

- *Bone* e.g., orthopedic plates & pins, bone prostheses,

- *Tissue and tissue fluid* (incl. prolonged mucous membrane) e.g., pacemakers, drug delivery devices, breast implants

- *Blood* e.g., pacemaker electrodes, vascular grafts, heart valves

FDA Biocompatibility Test Matrix

FDA Recommended Testing (1995)	II		III			IV			
	A	B	A	B	C	A	B	C	
Irritation tests	X	X	X	X	X	X	X	X	
Sensitization assay	X	X	X	X	X	X	X	X	
Cytotoxicity	X	X	X	X	X	X	X	X	
Acute systemic toxicity	X*	X	X	X	X	X	X	X	
Hemocompatibility		X	X	X	X	X	X	X	血液相容性
Pyrogenicity			X	X	X	X	X	X	致热
Implantation tests		X*	X	X	X	X	X	X	
Mutagenicity		X*	X*+	X	X	X	X	X	致突变性
Subchronic toxicity		X*+	X*+	X*+	X*+	X*+	X*+	X*+	
Chronic toxicity		X*	X*	X*	X*	X*	X*	X*	
Carcinogenesis bioassay			X*	X*	X*	X*	X*	X*	

+contact duration 5 min-29 days

*contact duration > 30 days

Tests Performed on Device Materials

Irritation tests: potential to cause skin/tissue irritation (animal or human)
Sensitization assay: potential for sensitization to a material (animal or human)
Cytotoxicity: cell growth inhibition or death (cell cultures)
Acute systemic toxicity: harmful effects of single or multiple exposures in < 24 hr period (animal)
Hemocompatibility: evaluation of hemolysis (red blood cell lysis), thrombosis, coagulation, platelet and immunological response
- Limited contact devices (in vitro)
- Extended contact with circulating blood (in vitro + animal model)

Pyrogenicity: material-mediated febrile response (animal)
Implantation tests: toxic effects on tissue at implant site (animal)
Mutagenicity: potential for gene mutations (cell cultures)
Subchronic toxicity: harmful effects of multiple exposures over short-term (1 day to 10% of animal lifetime e.g., 90 days for rat) (animal)
Chronic toxicity: harmful effects of multiple exposures over long-term (>10% of animal lifetime) (animal)
Carcinogenesis bioassay: tumorigenic potential over lifetime (2 yrs for rat) (animal)

IV catheters，静脉导管，Hemolysis, 溶血, Hemocompatibility,血液相容性;
Pyrogenicity, 制热性

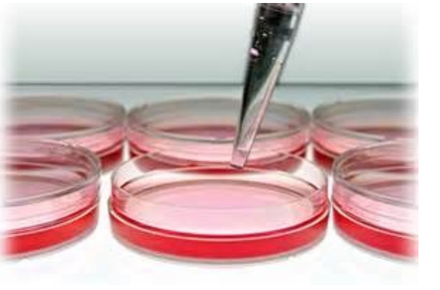
1. For in vitro and in vivo tests, we talked about 11 kinds of tests, list the names of four kinds of tests and the function or potential problems of one kind of test.

2. Implanted sites in animal models need test 4 basic types of tissues, list the names.

3. Three morphological cell culture assays are primarily used for evaluating biocompatibility, what are these assays?

In Vitro Testing
(cell cultures)

Used successfully to screen materials and devices for biocompatibility



- Advantages
  - Reasonable cost
  - Small to reasonable capital investments in lab and equipment
  - Relatively fast processing of large # of materials and prototypes

- Common test in blood compatibility of materials and devices using anticoagulated blood and evaluating formation of blood clots

In Vivo Testing
(animal experiments)

- Used to determine *in vivo* compatibility of materials and devices
  - Negative results mean system is unacceptable
  - Positive results do not necessarily prove compatibility in humans

- Should only be considered after successful completion of:
  - Prerequisite material characterization tests
  - Appropriate computer simulation models
  - Pertinent *in vitro* models

- better approximation to human environment
- demanding protocols (Animal Welfare Act)
- right animal model approximate human environment
- second step prior to clinical use

Cytotoxicity Tests

- Cytotoxicity means to cause toxic effects at the cellular level:

  - death,
  - alterations in cellular membrane permeability,
  - enzymatic inhibition,... at the cellular level.

- Evaluation by methods that use isolated, adherent cells in culture to measure cytotoxicity and biological compatibility.

  - Cells used for culture are most often established cell lines from cell banks (e.g. American Type Culture Collection)
  - Cultured cell lines can be reproducibly used in many different laboratories, providing comparable results useful for generating databases.

- Primary cells (with the exception of erythrocytes for hemolysis assays) are seldom used.

  Primary cells come directly from living tissue (can only be propagated a few generations in culture), have different genetic backgrounds, giving very high statistical variations in test outcome

Toxicity Tests

- A toxic material is defined as a material that releases a chemical in sufficient quantities to kill cells either directly of indirectly through inhibition of key metabolic pathways.

- The number of cells that are affected is an indication of the dose and potency of the chemical.

- If an animal is exposed to an atmosphere containing a noxious substance (**exposure dose**), only a small portion of the inhaled substance will be absorbed and delivered to the internal organs and cells (**delivered dose**).

- Cell culture methods evaluate target cell toxicity by using delivered doses of the test substance used. –Whereas tests in whole animals relate to the exposure dose... Often resulting in different measurements of sensitivity in the two systems. To compensate for this difference *in vivo* local toxicity models are applied (direct delivery to specific organs)

Toxicity Tests

A highly sensitive test system is desirable for evaluating the potential hazards of biomaterials.

- In cell culture the variables of metabolism, distribution, inflammation, and absorption are minimized and the dosage per cell is maximized to produce a highly sensitive test system. Testing at this high margin is considered a **safety factor** for interpolating results to whole humans

- Typical sources of toxic materials: extractables
  - Additives for manufacturability: plasticizers, antioxidants, monomers
  - Leakage from the basic material itself: cobalt, nickel from metal alloys; fluorinated polyesters from Dacron fibers; etc.

Toxicity Tests

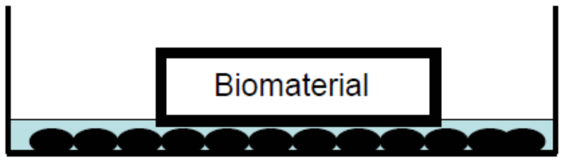
- Migration of chemicals from a solid phase material into liquid solvent is controlled by:
  - Diffusional resistance within the solid
  - Chemical concentration
  - Time，Temperature
  - Fluid turbulence at the solid-solvent interface

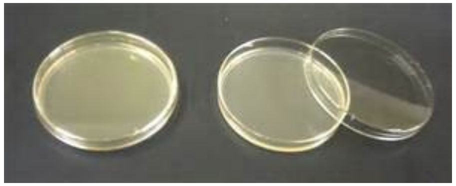
- Preparation of extractions of biomaterials have been carefully standardized to improve the reproducibility of the data.

- Complete dissolution of biomaterial is an alternative approach for in vitro testing. But:
  - May create degradation products that do not occur in the clinical application.

Toxicity Tests

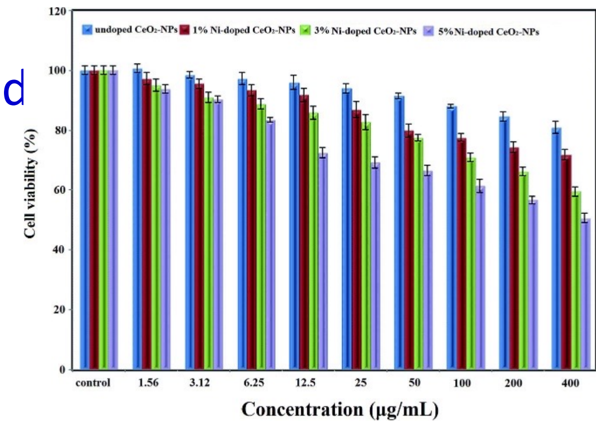
- Three morphological cell culture assays are primarily used for evaluating biocompatibility:
  - Direct contact
  - Agar diffusion
  - Elution



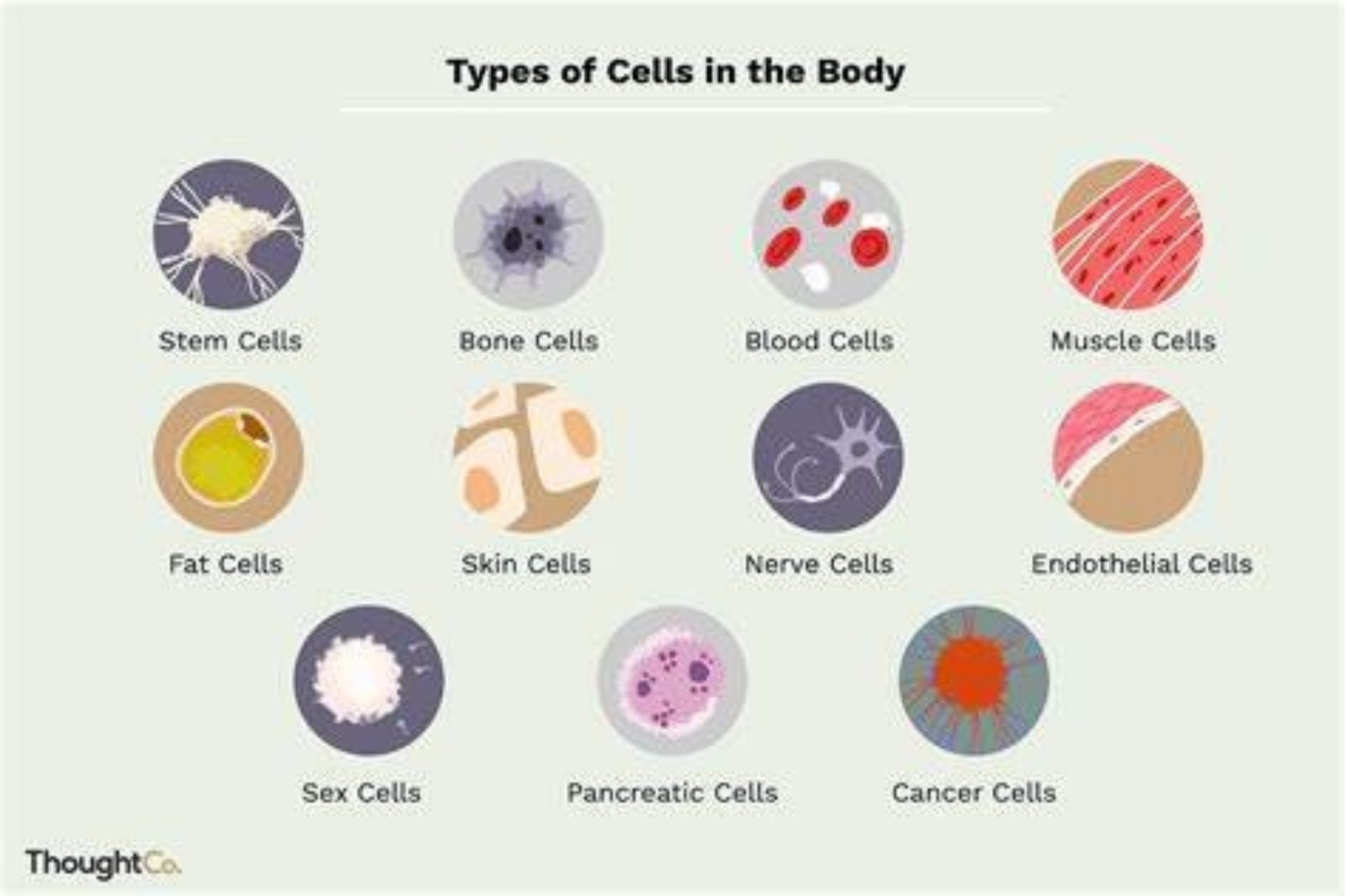


- For the results to be comparable the following parameters must be standardized:
  - Number of cells
  - **Growth phase of cells (self-learning if you have interests)**
  - **Cell type**
  - Duration of exposure
  - Test sample size (geometry, density, shape, thickness)
  - Surface area of test sample must be carefully controlled

- Readout: Within the given quantification range:
  - Dose-response curves
  - exposure-effect relationships (Klaassen, 1986)

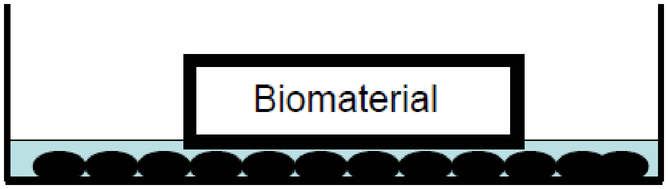


11 Different Types of Cells in the Human Body



Toxicity Tests

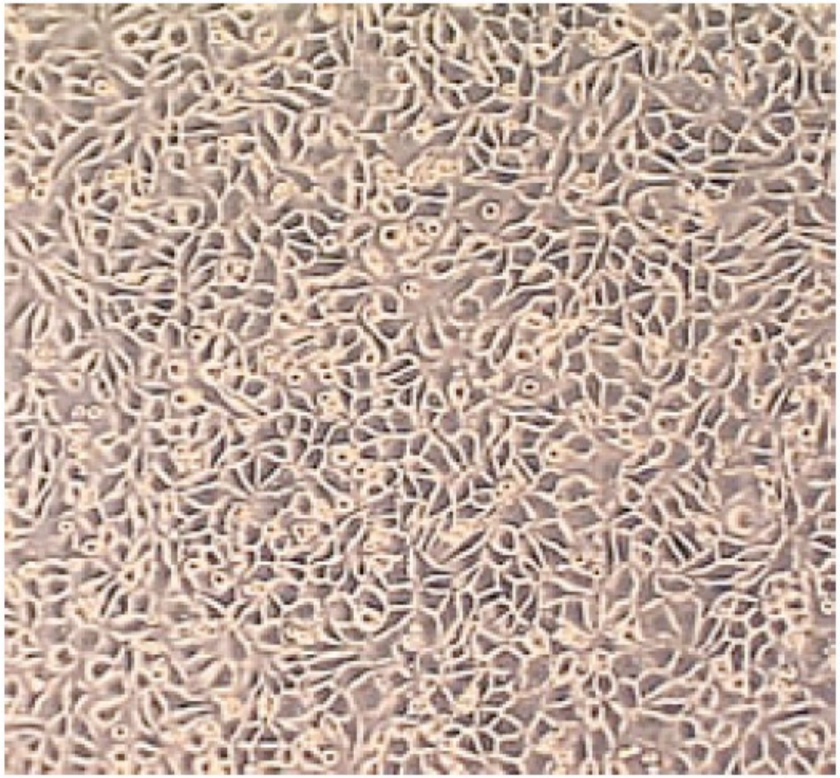
- **Direct contact**
  - monolayer, confluent cell culture, L-929 mouse fibroblasts
  - biomaterial in direct contact
  - 24 hours, 37±1°C
    - cells may change morphology
    - die
    - lose adherence to dish



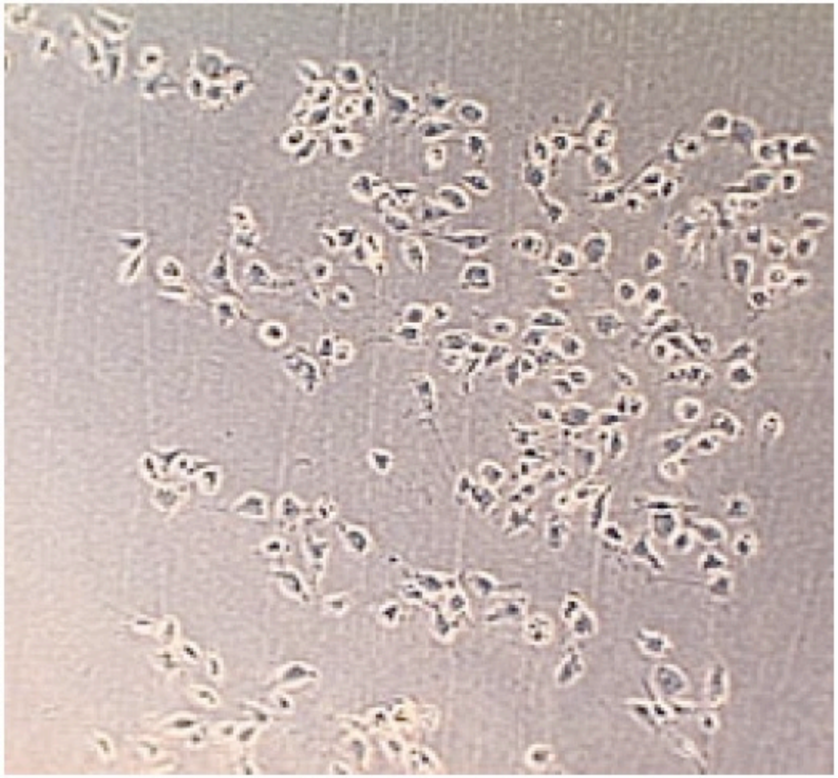
  - Cells are fixed and stained (hematoxylin blue: stains live adherent cells)
  - toxicity=dead/live !

- Why L-929?
  - easy to maintain
  - good correlation with animals tests (Northup 1986)
  - Resemble fibroblasts present in wound healing (=often the first cells to attach to implanted biomaterials *in vivo*)
  - In specific cases other similar cell types may be used

Direct Contact Observation



A confluent monolayer (100 x magnification) of well-defined L929 mouse fibroblast cells exhibiting cell-to-cell contact. This appearance is indicative of a non-cytotoxic (negative) response



L929 mouse fibroblast cells (100 x magnification) that illustrate a positive cytotoxic reaction; the considerable open areas between cells indicate that extensive cell lysis (disintegration) has occurred.

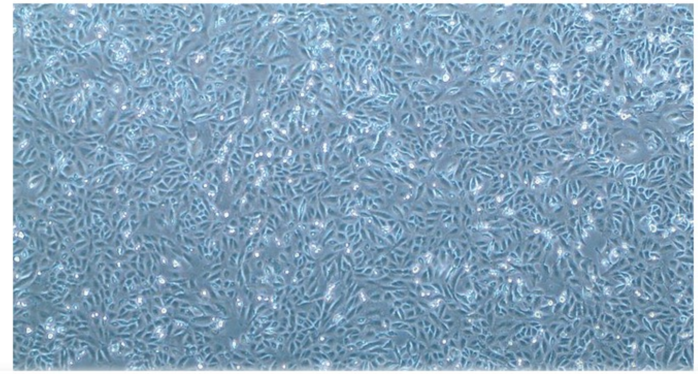
Cell confluency

In cell culture biology, confluency is commonly used as a measure of the number of the cells in a cell culture dish or a flask and refers to the coverage of the dish or the flask by the cells. For example, 100 per cent confluency means the dish is completely covered by the cells and there are no more room left for the cells to grow where as 50 per cent confluency means roughly half of the dish is covered and there is room for cells to grow.

另外, cell confluency除了表示细胞密度以外，还综合反映了细胞生长过程的一种状态。

因为细胞在体外培养生长时具有一些特点，尤其是贴附和接触抑制及密度依赖性。一般来说，当细胞生长、汇合形成单层时，细胞变得比较拥挤，而扁平形状的程度减小，与培养液接触的表面区域亦因而减少；同时培养液中的一些营养物将逐渐消耗掉，尤以紧靠着细胞周围的部位。这种形成单层的细胞，其分裂将停止。所以汇合度的概念并不仅指细胞密度，它是综合反映了细胞生长过程的一种状态。

Snu449 cells at about 100 per cent confluency.



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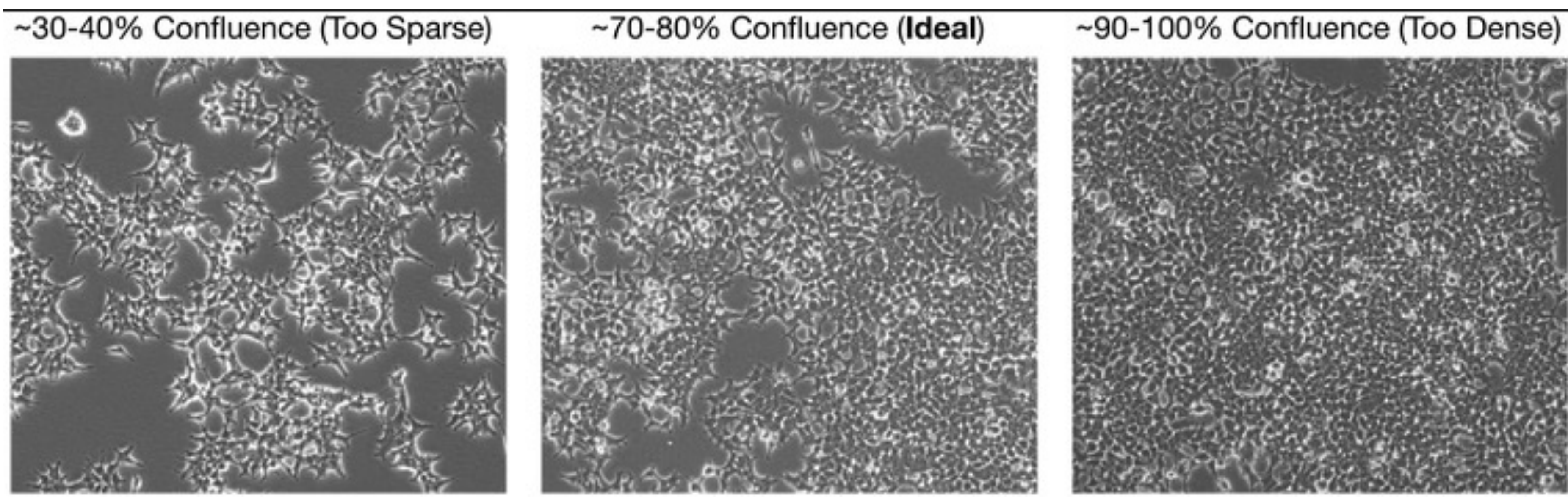
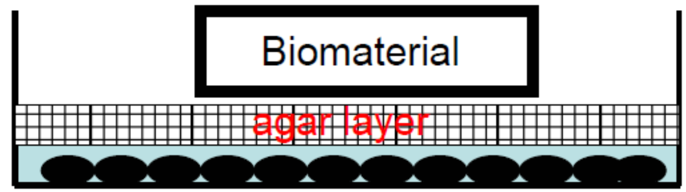
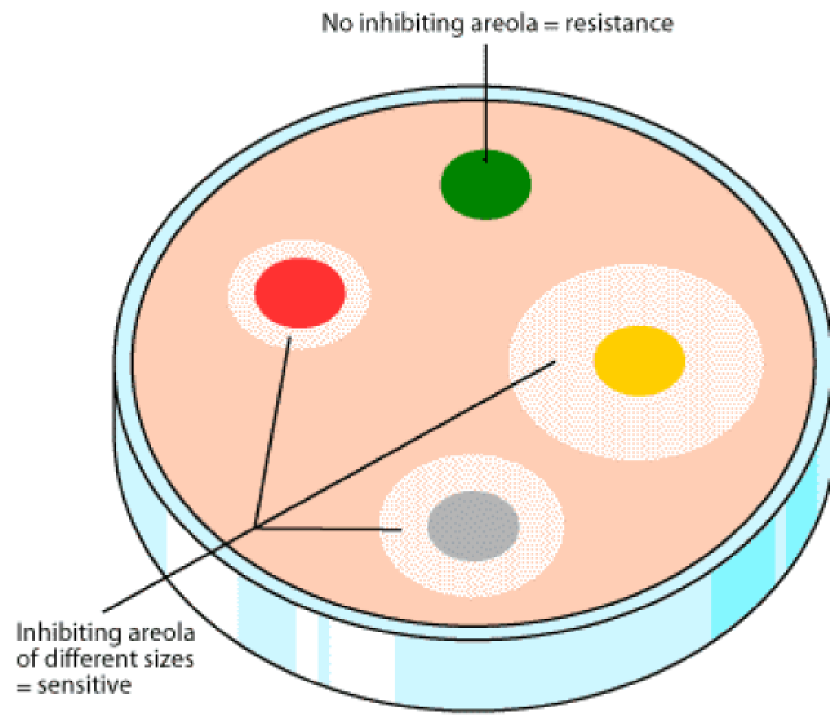


Agar Diffusion

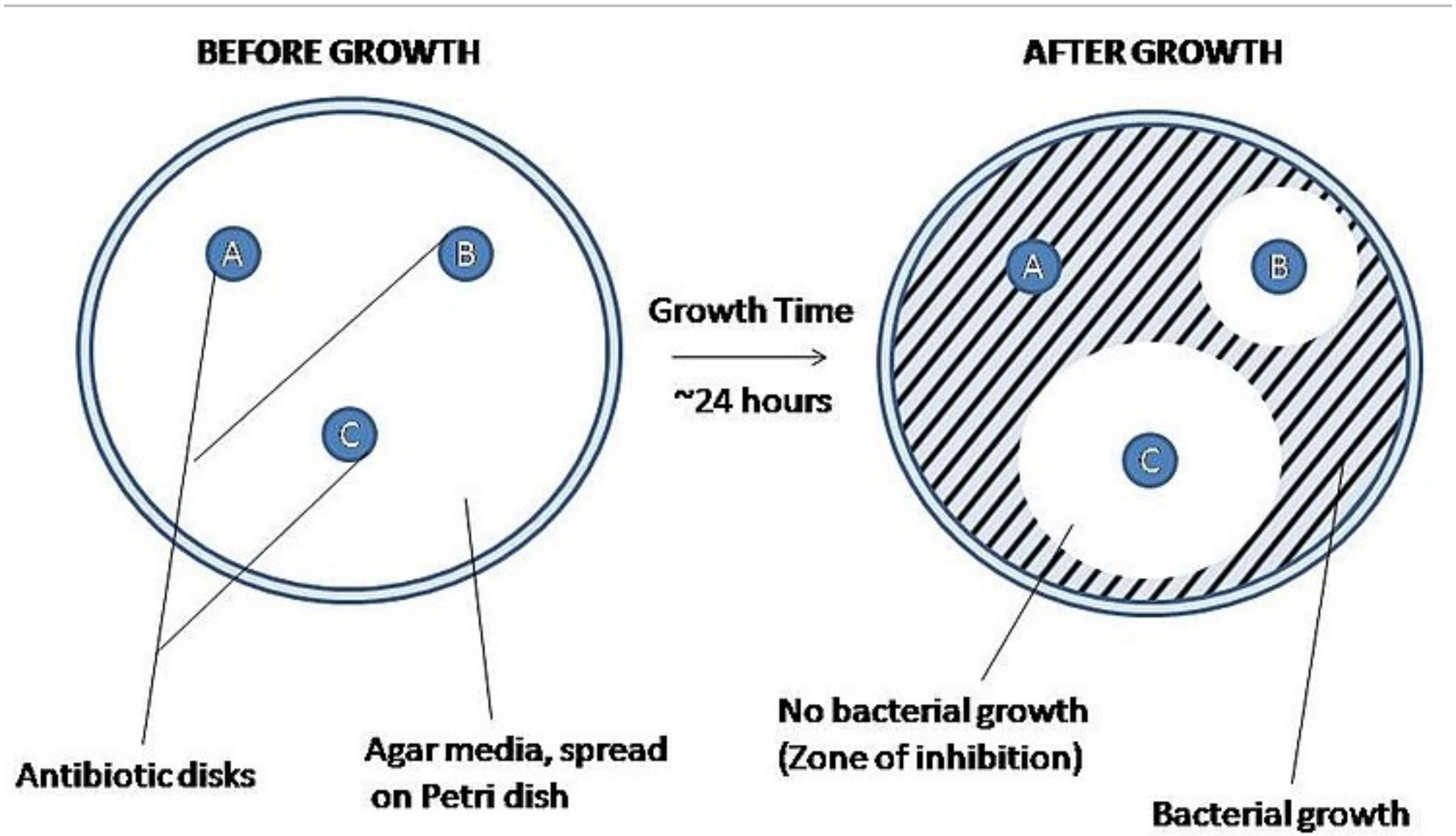
- Agar layer between cells and biomaterial
- agar: gel-like polymer derived from red alga
- chemicals diffuse through agar
- Death of injured cells areas remain colorless





洋菜，亦称**琼脂**、**琼脂胶**、**大菜**、**菜燕**，是从海藻类植物中提取的胶质。1660年代由日本美浓屋的太郎左卫门提取出来。可作为鱼胶的代用品，常被用于沙拉、大菜糕或果冻等甜品。
琼脂亦会使用于实验室，通常作为细菌的培养媒体或用于分子生物学实验

Agar Diffusion



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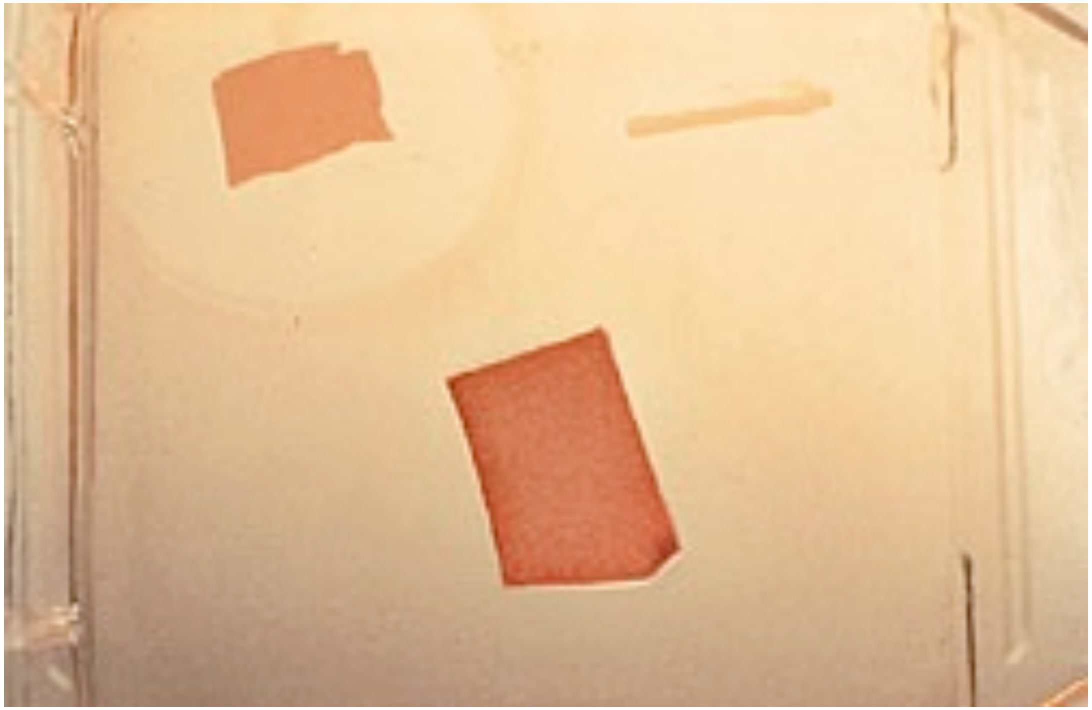
https://commons.wikimedia.org/wiki/File:Agar_Diffusion_Method_1.jpg

Agar Diffusion



An agar diffusion flask containing a sample of positive control material. The discoloration that extends outward from the material indicates that the presence of the sample has caused the cells to lyse.

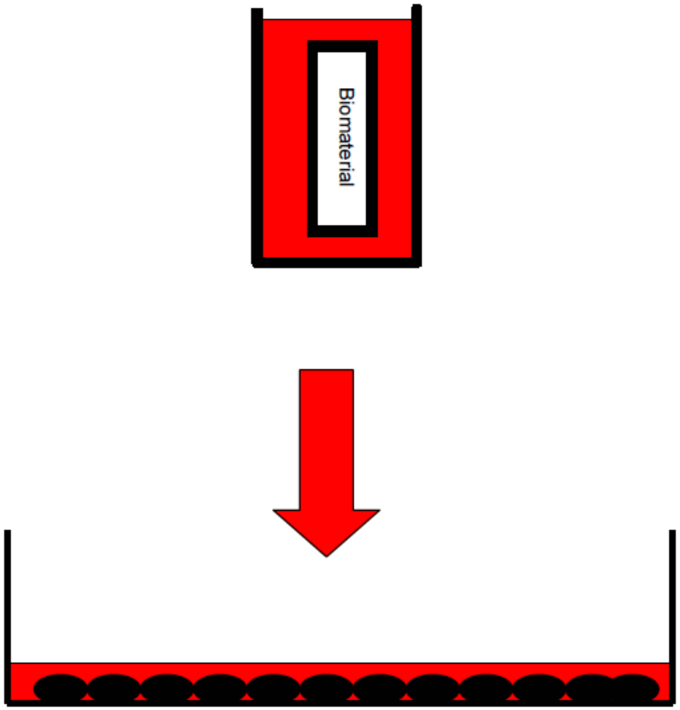
Difference-between-agar-and-agarose

Red algae or seaweeds are mainly used to produce different types of polysaccaradises such as agar and agarose. This seaweed is commonly cultivated in some parts of Asia and the United states. Agar and agarose are used in different applications such as culturing micro-organisms and as a culinary ingredient. The terms agar and agarose are frequently used interchangeably since they are closely interconnected. However, there is a difference; **Agarose is derived by purifying agar. In contrast, agar is directly derived from red algae.** This is the main difference between agar and agarose. Agar is also cheaper than agarose. Due to gel-like characteristics, both of these materials are used in the field of microbiological analysis and deliver nutrients to microorganisms. Furthermore, agar is commonly used in the food industry as a food ingredient whereas agarose is ordinarily used in gel electrophoresis. In this article, let’s look at the difference between agar and agarose in terms of their physical, chemical characteristics and intended uses.

https://pediaa.com/difference-between-agar-and-agarose/

Elution

- prepare extract of a material
– how? Standardization needed! (0.9% sodium chloride or serum free medium)
– chemicals will leak into solution
– apply solution to cell-culture (48h at 37°C)
– perform stain based viability tests, microscopic evaluation.
– experience in recognizing cell culture morphology is required.



Comparison of the Three Cell Culture Methods

TABLE 1 Advantages and Disadvantages of Cell Culture Methods

	Direct contact	Agar diffusion	Elution
Advantages	Eliminate extraction preparation	Eliminate extraction preparation	Separate extraction from testing
	Zone of diffusion	Zone of diffusion	Dose response effect
	Target cell contact with material	Better concentration gradient of toxicant	Extend exposure time
	Mimic physiological conditions	Can test one side of a material	Choice of extract conditions
	Standardize amount of test material or test indeterminate shapes	Independent of material density	Choice of solvents
	Can extend exposure time by adding fresh media	Use filter paper disk to test liquids or extracts	
Disadvantages	Cellular trauma if material moves	Requires flat surface	Additional time and steps
	Cellular trauma with high density materials	Solubility of toxicant in agar	
	Decreased cell population with highly soluble toxicants	Risk of thermal shock when preparing agar overlay	
		Limited exposure time	
		Risk of absorbing water from agar	

Testing Biomaterials

- After the cytotoxicity profile more application-specific tests are performed to assess the biocompatibility of the material:

  - Products for ***in vitro*** **fertilization** procedures would be tested for adverse **effects on a very low cell population**.

  - A new **material for culturing cells** would be assayed by comparing growth rates of cells in contact with the new material with those of **currently marketed materials.**

  - Current experience: a material **non-toxic *in-vitro*** will be **non-toxic in *in-vivo*** assays.
  - **But:** the **clinical acceptability** of a material depends on **many different factors**; target cell toxicity is but one.

In Vivo Testing

In vivo testing: critical for development of clinical devices

- *In vitro* tests cannot replace *in vivo* tests:
  - no inflammation
  - No immune response
  - single cell type
  - no tissue remodeling
  - No acquired toxicity through processing (e.g. the liver modifies many foreign compounds)

In vivo tests provide:
- interactions of different cell types
- effects of hormonal factors
- interactions with extracellular matrix
- interactions with blood-borne cells, proteins and molecules
- Overall determination of: whether the device performs as intended and provides no significant harm to the patient or user.

The ISO 10993 standard, "Biological evaluation of Medical Devices" presents a systematic approach to the *in vivo* assessment of tissue compatibility of medical devices. The Standard is extended and upgraded continuously.

In Vivo Testing

Implant effects can be simulated *in vivo*:

- insoluble particulate materials released by implants
- interaction of biological factors with the implant
- mechanical loading experienced by device
- Time is an important variable (implant–related factors act with different time constants on the biological factors)

- The tissue response to an implant is the cumulative physiological effect of:
  - Modulation of the acute wound healing response due to the surgical trauma of implantation and the presence of the implant.
  - The subsequent chronic inflammation reaction, and
  - Remodeling of surrounding tissue as it adopts to the implant.

Mechanical Loading Experienced by Biomaterial

- increased local strain due to movement of device with respect to tissue:
  hyperplasia (increased scar tissue, thicker fibrous encapsulation)

- reduction in tissue strain due to presence of implant implant takes all load:
  tissue undergoes atrophy (*stress shielding*)

Figure 1. Example of stress shielding. (A: postoperative image. B: image of the same patient after 7 years of follow-up. Arrow indicates stress shielding)

Testing Biomaterials

Implant sites in animal models:

- Similarity to the site to be employed in human use of the medical device.
- The healing and remodeling characteristics of the 4 basic types of tissue should be considered:
  - Connective tissue
  - Muscle
  - Epithelia
  - Nerve

- In selecting an implant site consider the following:
  - Vascularity
  - Nature of the parenchymal cells (capability for mitosis and migration: determine the regenerative capability of the tissue)
  - Presence of regulatory cells (macrophages and histiocytes)
  - Effects of mechanical strain (hyperplasia, atrophy)

parenchymal cell: 实质细胞。一个器官内，承担该器官功能的细胞

Histiocyte: 组织细胞网络组织球；组织细胞学检查；组织球细胞

Testing Biomaterials

For orthopedic prostheses bone has been used as the site of implantation:

- But the density of bone formation depends on the site of implantation:
  - Cortical and cancellous bone differ in vascularization and the size of the pool of preosteoblasts that proliferate in response to surgery.

Cutaneous or subcutaneous sites chosen to assess biocompatibility

- readily accessible
- thickness of fibrous capsule measure of biocompatibility
- guinea pig

Preosteoblasts，胚胎前成骨细胞网络骨母前趋细胞；骨前趋细胞；前造骨细胞

Testing Biomaterials

- Materials for vascular prostheses have been evaluated for their blood compatibility as replacements segments in selected vessels in various animal models:

  - Carotids-jugular and （颈总动脉)

  - Femoral arteriovenous shunts (颈静脉和-股动静脉分流术)

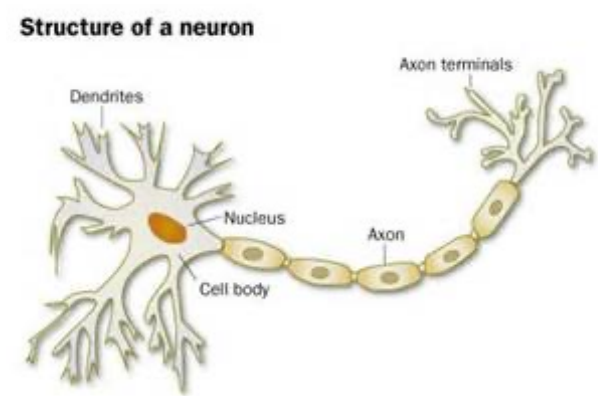
- Nerve:

  - Nerve cells do not have the capability for division

  - The elongation of several axons (轴突) allow a degree of regeneration across defect sites.

  - Certain matrices facilitate the elongation of such axons thereby accelerating the regeneration of the nerve and restoration of some function:

    Peripheral nerves of rats



Testing Biomaterials

Evaluation of tissue reaction:

- Histology and histochemistry:

  Qualitative determination of the relative numbers of various cell types.

- Immunohistochemistry:

  Allows specific cell types and extracellular matrix components around an implant to be identified.

- Transmission electron microscopy (TEM):

  Ultrastructural examination of cells at the interface of implants.

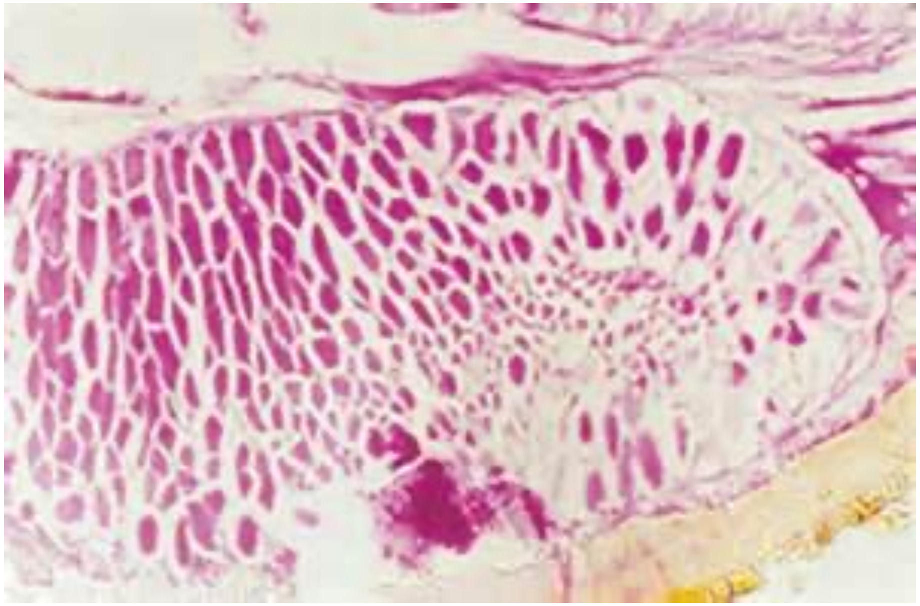
- Scanning electron microscopy (SEM)

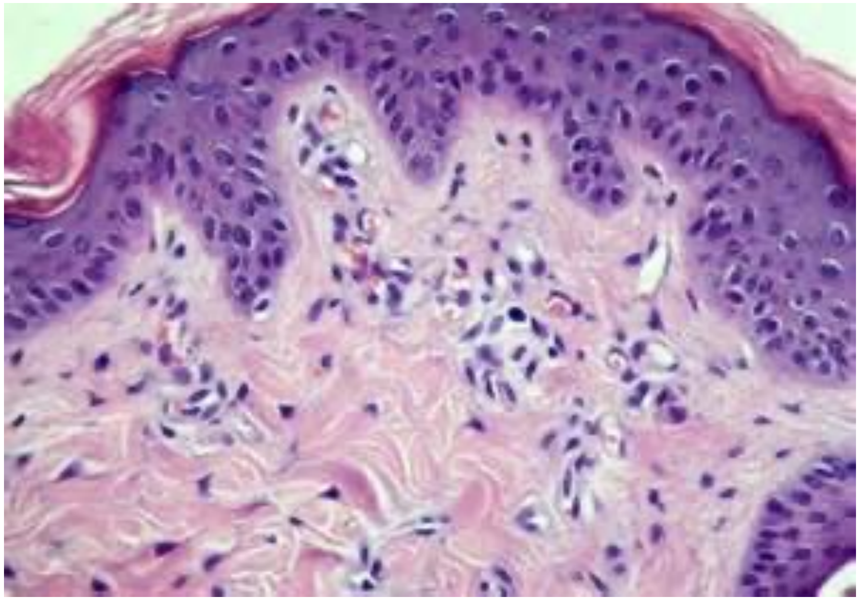
- Biochemistry:

  Level of inflammatory mediators

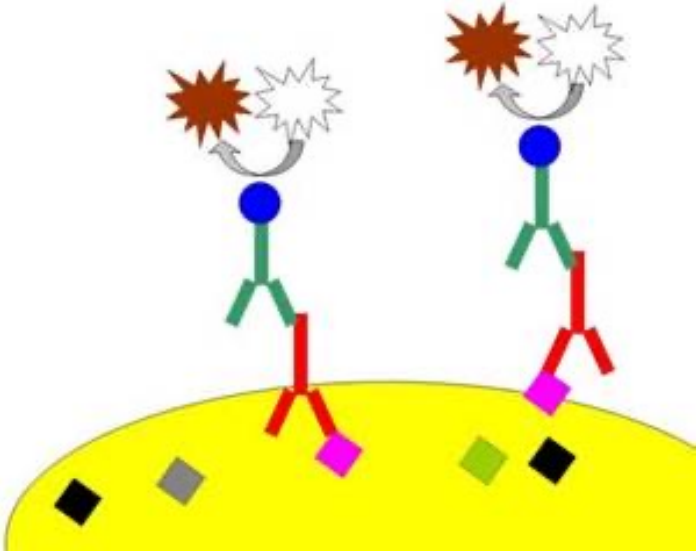
  But: Manipulation of tissues at or after explantation of a biomaterial can dramatically alter the production and release of cellular mediators.

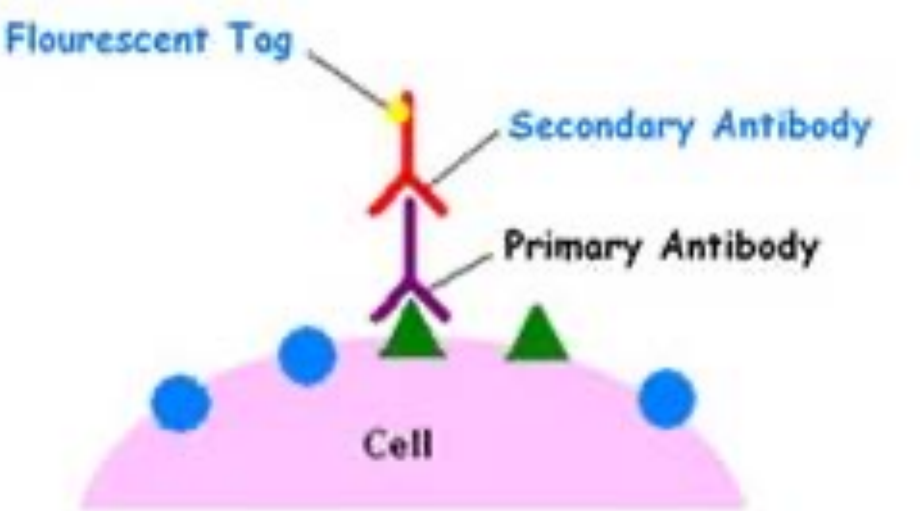
Histology and histochemistry:





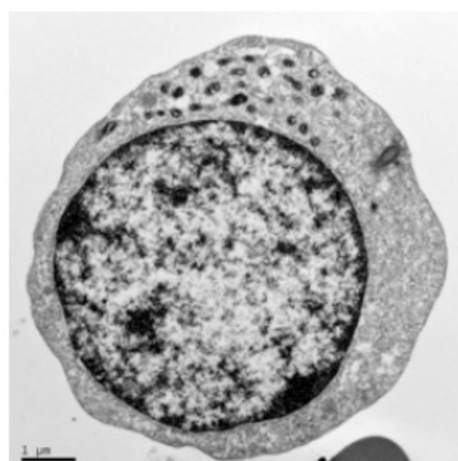
Immunohistochemistry





A typical procedure for cell fixation and cells on nanofiber

1. Rinse any residual protein off of surface with PBS 1 X 5 min (optional)
2. Fix cells in 4% paraformaldehyde (or 10% formalin) in PBS for 30 minutes at room temperature
3. Rinse with PBS 1 X 5 minutes
4. Dehydrate in graded ethanol series:
    a. 50% EtOH balance with distilled water 1 X 5 min
    b. 70% EtOH balance with distilled water I X 5 min
    c. 80% EtOH balance with distilled water 1 X 10 min
    d. 95% EtOH balance with distilled water 1 X 10 min
    e. 100% EtOH 2 X 10 min
5. Hexamethyldisilazane (HMDS) chemical drying series
    a. 3:1 EtOH:HMDS 1 X 15 min
    b. 1:1 EtOH:HMDS 1 X 15 min
    c. 1:3 EtOH:HMDS 1 X 15 min
    d. 100% HMDS allow to air dry
6. Mount on SEM stub and sputter coat



Testing of Blood-Material Interactions

- Many devices and materials used have blood contact:
  - Heart-lung machine
  - Hollow fiber hemodialyzer for treatment of kidney failure
  - Catheters for blood access and
  - Blood vessel manipulation (angioplasty)
  - Heart assist devices
  - Stents
  - Prosthetic heart valves
  - Vascular grafts

- A device made of blood-compatible materials is not automatically blood compatible!
- Not widely recognized, standard list of blood compatibility tests exists.

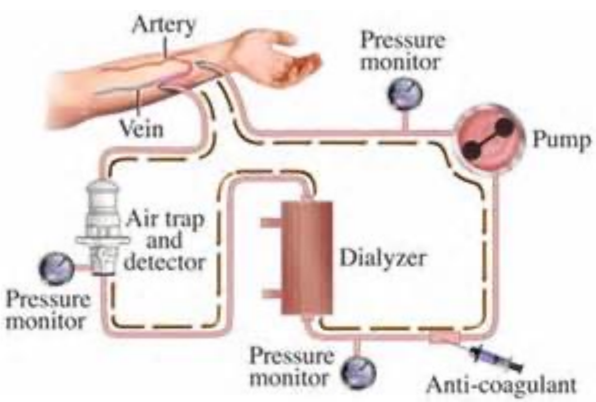
Testing of Blood-Material Interactions

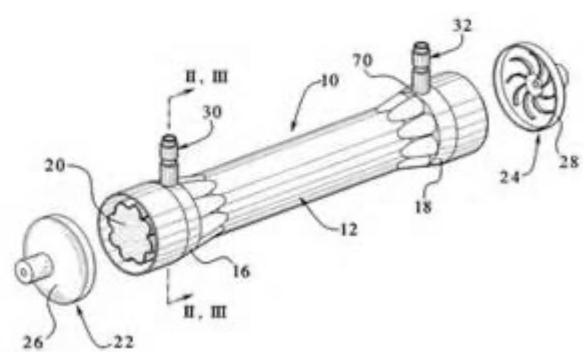
- Many existing devices are frequently modified to improve durability and mechanical characteristics.

  - Changes may also affect blood response (is not entirely predictable): testing is required to document safety

- The performance of many existing devices is also less than optimal:

  - Prolonged heart lung machine can produce a tendency to severe bleeding.
  - Mechanical heart valves occasionally shed emboli to the brain, producing stroke.
  - Many devices are only “safe” when anticoagulating drugs are used (e.g. Oxygenator, heart valves, hemodialyzer (血液透析器人工肾)).





Thrombogenicity (致血栓形成能力；致血栓性；凝血活性)

Local effects:

- A thrombogenic device may cause the accumulation of various blood elements (thrombus formation).
- Cardiovascular devices may also exhibit regions of disturbed flow or stasis which lead to formation of blood clots.
- These local effects can compromise device function:
  - Delivery of blood through artificial blood vessels
  - Mechanical motion of heart valves
  - Gas exchange through oxygenators
  - Removal of metabolic waste (hemodialyzer)

Thrombogenicity

The local blood reaction may produce systemic effects:

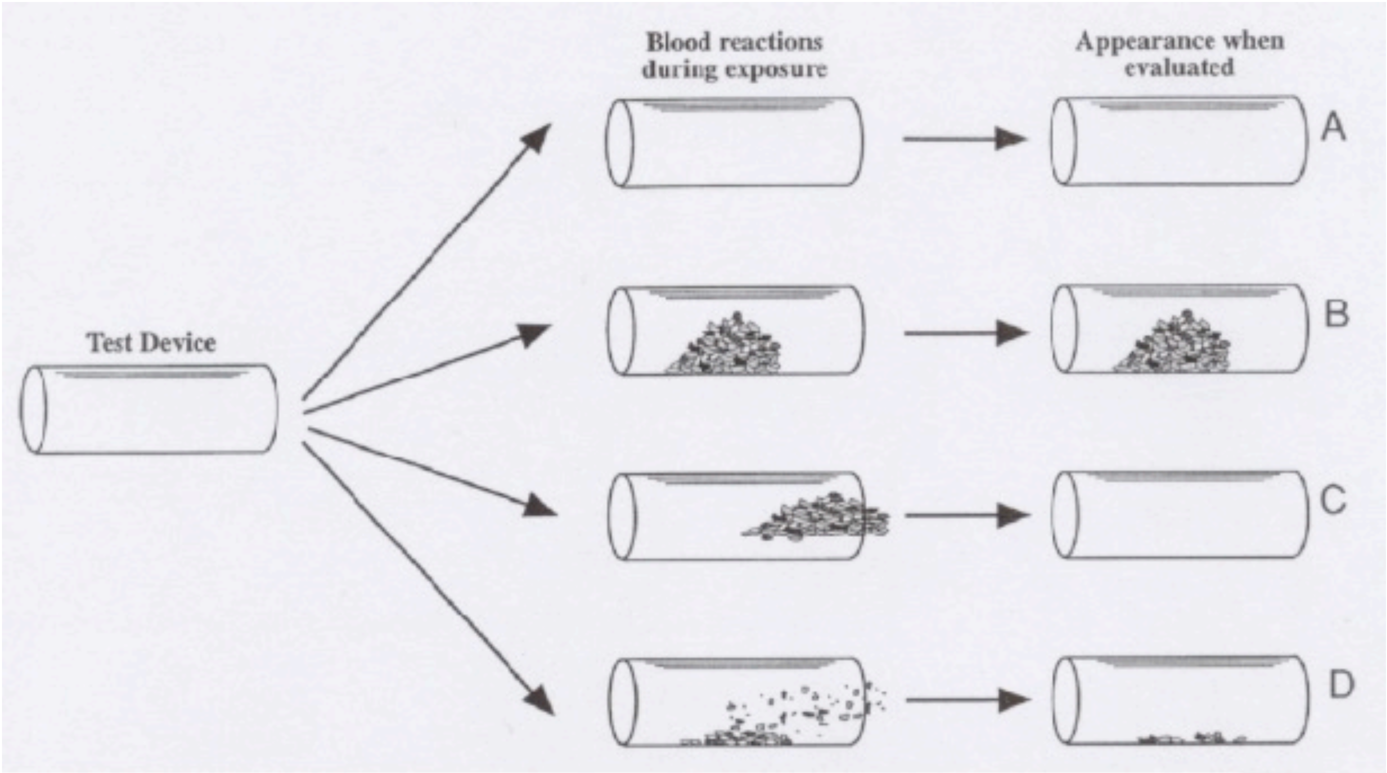
- Thrombi may detach (embolize, 栓塞) and impair blood flow in peripheral vessels.

- Chronic devices may “consume” circulating blood elements:
  - Mechanical destruction of red blood cells by heart prostheses or dialyzers.
  - Removal of platelets as a result of continuing thrombus formation.
  - Mediators of inflammatory responses and vessel tone may be produced or released from cells (platelets, white cells,..).

Thrombogenicity

- The types of devices used are:
  - numerous, exhibit complex flow geometries, and are continuously evolving.
- The possible blood responses are:
  - numerous, complex, dynamic, and not fully understood.

- It is difficult and expensive to measure device thrombogenicity in an extensive and systematic way (experiment. animals or humans).
- Alternative interpretations can be applied to data from “blood compatibility” tests.



Quiz:

1. Compare the advantages and disadvantages between in-vivo and in-vitro assays (list 3 points each)

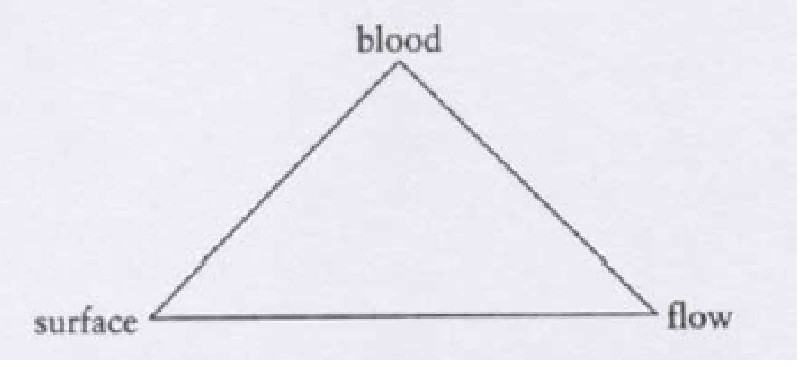
2. For toxicity test, the standard method uses L-929 mouse fibroblast cell lines, what are the reasons? (list at least 2 points)

Testing of Blood-Material Interactions

- We cannot generally:

  - Extrapolate results obtained under one set of test conditions to another set of conditions.
  - Use short-term testing to predict long-term results.
  - Predict in vivo performance of a device based on blood-material interactions of materials per se in idealized flow geometries.

- 3 factors contribute to the coagulation of the blood:
  - The blood chemistry
  - The blood-contacting surface
  - The flow regime



Virchow´s triade (1856)

Extrapolate results： 推断结果

Testing of Blood-Material Interactions

- The source and methods for handling blood can have important effects on blood material interactions.
- Initial adhesiveness of blood platelets for artificial surfaces appears to be
  - low in man and some primates and
  - high in the dog, rat and rabbit.
- Animal blood donors are relatively homogenous:
  - Age, health status, blood response
- In vitro testing generally requires anticoagulation of the blood (can have profound effects).
- In vivo testing and the use of extracorporeal circuits are also commonly performed with anticoagulats:
  - Sodium citrate (chelates Ca^{2+})
  - Heparin (used to block thrombin)

Extracorporeal：体外

Testing of Blood-Material Interactions

TABLE 2 Blood–Materials Responses and Their Evaluation

System	Blood response	Assessment[a]
Device/material	Thrombosis	Direct visual and histologic evaluation; noninvasive imaging (angiography, ultrasound, radioisotope, magnetic resonance); evidence of device dysfunction.
	Thromboembolism	Emboli detection (ultrasound, laser); evidence of organ/limb ischemia, stroke.
Platelets	Consumption	Increased removal of radioisotopically labeled cells; reduced blood platelet count.
	Dysfunction[b]	Reduced platelet aggregation *in vitro;* prolonged bleeding time.
	Activation	Increased plasma levels of platelet factor 4 and β-thromboglobulin; platelet membrane alterations (e.g., by flow cytometry).
Red cells[b]	Destruction	Decreased red cell count; increased plasma hemoglobin.
White cells[b]	Consumption/activation	Decreased counts of white cell populations; increased white cell plasma enzymes (e.g., neutrophil elastase).
Coagulation factors	Consumption[b]	Reduced plasma fibrinogen, factor V, factor VIII.
	Thrombin generation	Increased plasma levels of prothrombin fragment 1.2 and thrombin : antithrombin III complex.
	Fibrin formation	Increased plasma level of fibrinopeptide A.
	Dysfunction[b]	Prolonged plasma clotting times.
Fibrinolytic proteins	Consumption[b]	Reduced plasma plasminogen level.
	Plasmin generation	Increased plasma level of plasmin : antiplasmin complex.
	Fibrinolysis	Increased plasma level of fibrin D-dimer fragment.
Complement proteins[b]	Activation	Increased plasma levels of complement proteins C5a and C3b.

[a]Radioimmunoassays (RIA) and enzyme-linked immunoassays (ELISA) may not be available for detection of nonhuman proteins.
[b]Tests which may be particularly important with long-term and/or large surface area devices.

Testing of Blood-Material Interactions

In vitro tests:

- Usually of short duration
- Strongly influenced by the blood source, handling methods, the use of anticoagulants
- Can not predict longer term BMI and in vivo outcome events
- Useful in screening materials

In vivo tests:

- Insertion for short/long periods into the arteries or veins of experimental animals.

  Arteriovenous (AV) or

  Arterio-arterial (AA) shunt

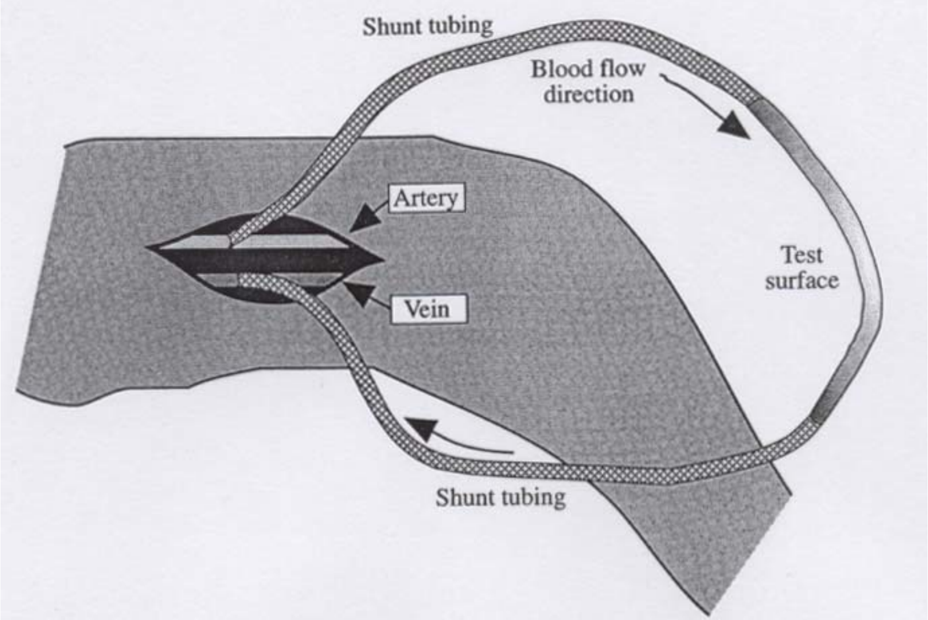


FIG. 3. Illustration of an arteriovenous (A-V) shunt placed between the femoral artery and vein (in the leg) of an experimental animal. Materials to be tested (in this case a tubular device) are interposed between inlet and outlet segments of the shunt.

动静脉的/动脉动脉瘘

Testing Biomaterials

Sensitization:

- Prolonged contact with a chemical substance that interacts with immune system
- Skin widely used since most reactions to biomaterials are cell-mediated type
- Dermal sensitization

marked by redness

and swelling

Sensitization (rash) to latex gloves

Testing Biomaterials

Sensitization test methods (guinea pigs):

- repeated patch (Buehler):

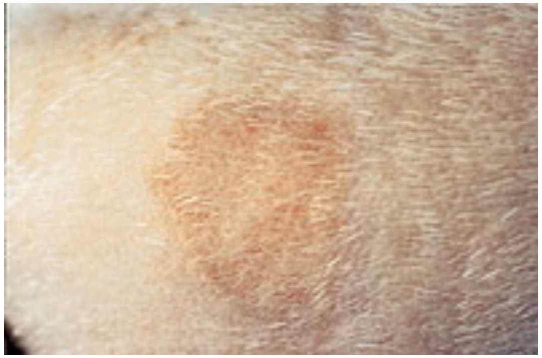
  - Induction phase: expose shaved back directly to material under occlusive dressings. 6 hours/day, 3 days/week, 3 weeks

  - Recovery phase: 2 weeks rest to allow for development of response

  - Final exposure

- maximization (Magnuson-Kligman): used for materials that will contact areas other than the skin:

- fluid extracts of test material prepared in saline or vegetable oil
- inject extract with an adjuvant agent that will enhance immune response
- two weeks rest
- apply extract topically



Positive maximization test in guinea pig

Testing Biomaterials

- **Irritation:** local tissue response characterized by the usual signs of inflammation:
  - redness
  - swelling
  - heat
  - pain

- In vivo tests for irritation:
  - intracutaneous
  - primary skin
  - ocular

Testing Biomaterials

Intracutaneous test:

- albino rabbits
- prepare fluid extract under controlled temperature, duration, material surface/volume ratio (water and oil based solvent)
- extract injected into multiple sites the skin (+ control injections)
- observe for evidence of redness and swelling at 24h, 48h, 72h

- aggressive test, extract pre-pared under exaggerated conditions

  maximizes the chance of finding irritant chemical

Intracutaneous irritation test using albino rabbits

Testing Biomaterials

Primary skin test

- less aggressive than intracutaneous
- placement of material on shaved back of albino rabbits
- cover with occlusive dressing
- apply between 4-24 hrs
- observe for 72 hrs
- score for redness and swelling
- compare with known values for primary skin irritation
- categorize the response: negligible, slight, moderate, severe

Testing Biomaterials

Ocular test:

- used for eye contact products
- fluid extracts (occasionally solids or powders)
- placed directly into the pocket of the lower eyelid of an albino rabbit
- other eye untreated, control
- observe regularly up to 72 hours
  - score based on: swelling and redness of conjunctiva (结膜)
  - response of iris to light
  - corneal opacity
  - presence of discharge

Testing Biomaterials

Systemic effects:

- Effects of released chemicals on liver, heart, kidneys, and brain
- Mice and rats; Various routes of application
  - dermal
  - inhalation
  - Intravenous (静脉)
  - Intraperitoneal (腹腔内)
  - Oral
- Application
  - fluid extracts (intraperitoneal or intravenous)
  - implantation of material (particularly biodegradable ones) (intramuscular, intraperitoneal, subcutaneous)
- Collect
  - blood samples (hematology, serum chemistry)
  - tissue samples (pathology)

Testing Biomaterials

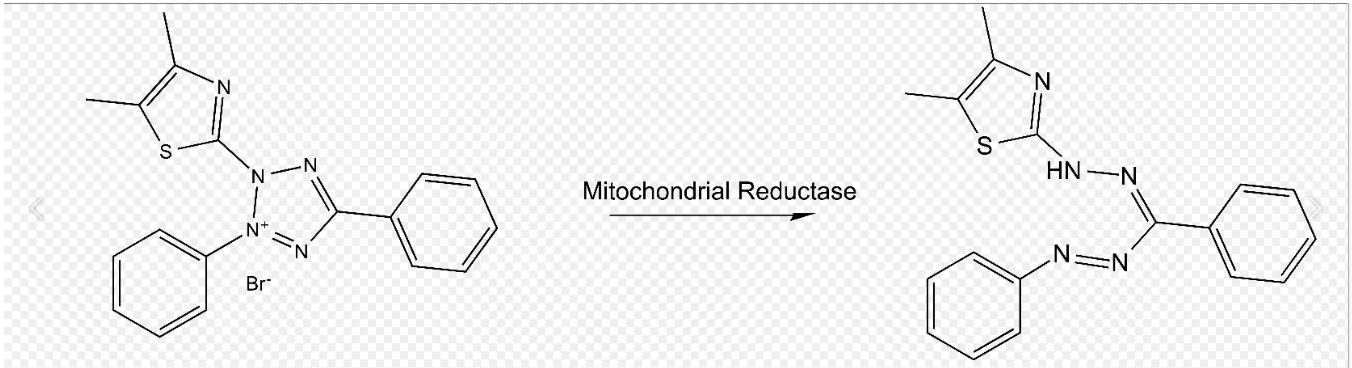
- **A hierarchy of testing**, starting with in vitro systems and progressing through functionality implants in situ is implied.

Cell culture cytotoxicity (mouse L929 cell line)
Hemolysis (rabbit or human blood)
Mutagenicity (human or other mammalian cells or Ames test (bacterial)
Systemic injection acute toxicity (mouse)
Sensitization (guinea pig)
Pyrogenicity (limulus amebocyte lysate [LAL] or rabbit)
Intracutaneous irritation (rabbit)
Intramuscular implant (rat, rabbit)
Blood compatibility (rat, dog, primate, etc.)
Long-term implant (rat)

Cell Viability & Proliferation Assays

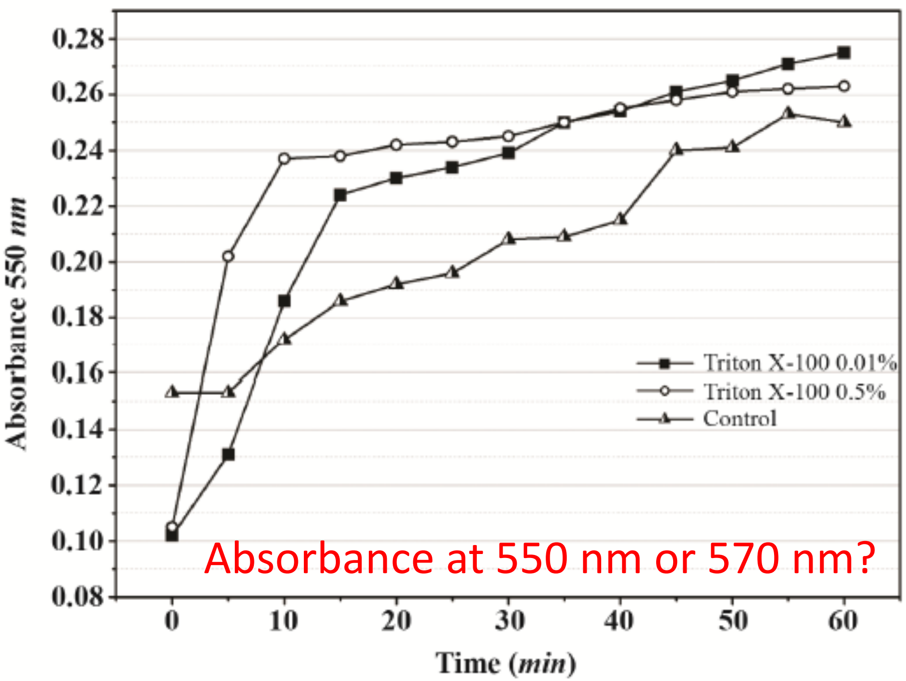
MTT assay

Viability assay

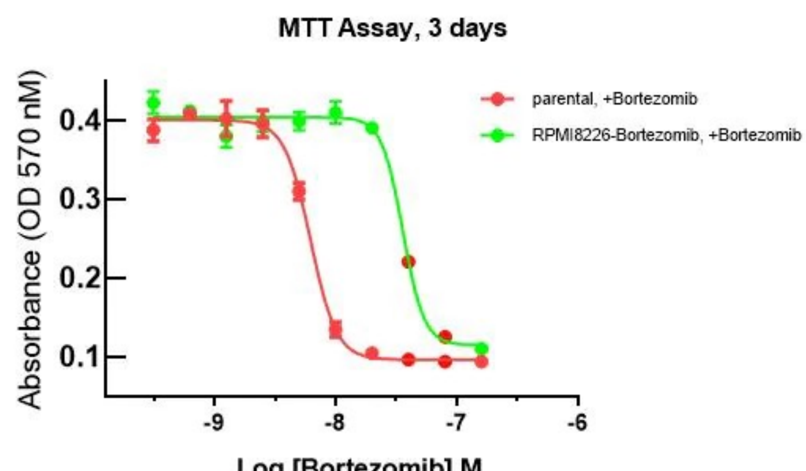


3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
(MTT)

(*E,Z*)-5-(4,5-dimethylthiazol-2-yl)-1,3-diphenylformazan
(Formazan)



Absorbance at 550 nm or 570 nm?



	parental, +Bortezomib	Bortezomib, +Bortezomib
IC50	6.198e-009	3.588e-008

MTT assay

其检测原理为活细胞线粒体中的琥珀酸脱氢酶能使外源性MTT还原为水不溶性的蓝紫色结晶甲瓒（Formazan）并沉积在细胞中，而死细胞无此功能。二甲基亚砜（DMSO）能溶解细胞中的甲瓒，用酶联免疫检测仪在540 或720nm波长处测定其光吸收值，可间接反映活细胞数量。在一定细胞数范围内，MTT结晶形成的量与细胞数成正比。该方法已广泛用于一些生物活性因子的活性检测、大规模的抗肿瘤药物筛选、细胞毒性试验以及肿瘤放射敏感性测定等。它的特点是灵敏度高、经济。

Why 540 nm and 720 nm??

用酶标仪在490nm波长处测定其光吸收值，在一定细胞数范围内，MTT结晶形成的量与细胞数成正比。根据测得的吸光度值(OD值)，来判断活细胞数量，OD值越大，细胞活性越强(如果是测药物毒性，则表示药物毒性越小)。

The MTT assay is a colorimetric assay for assessing cell metabolic activity. NAD(P)H-dependent cellular oxidoreductase enzymes may, under defined conditions, reflect the number of viable cells present. These enzymes are capable of reducing the tetrazolium dye MTT 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide to its insoluble formazan, which has a purple color. Other closely related tetrazolium dyes including XTT, MTS and the WSTs, are used in conjunction with the intermediate electron acceptor, 1-methoxy phenazine methosulfate (PMS). With WST-1, which is cell-impermeable, reduction occurs outside the cell via plasma membrane electron transport.[1] Tetrazolium dye assays can also be used to measure cytotoxicity (loss of viable cells) or cytostatic activity (shift from proliferation to quiescence) of potential medicinal agents and toxic materials. MTT assays are usually done in the dark since the MTT reagent is sensitive to light.

MTT assay

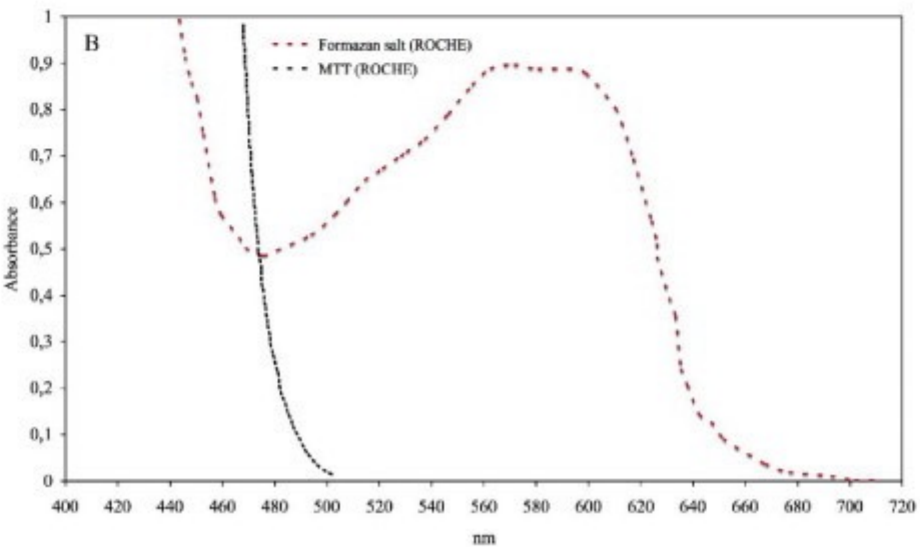
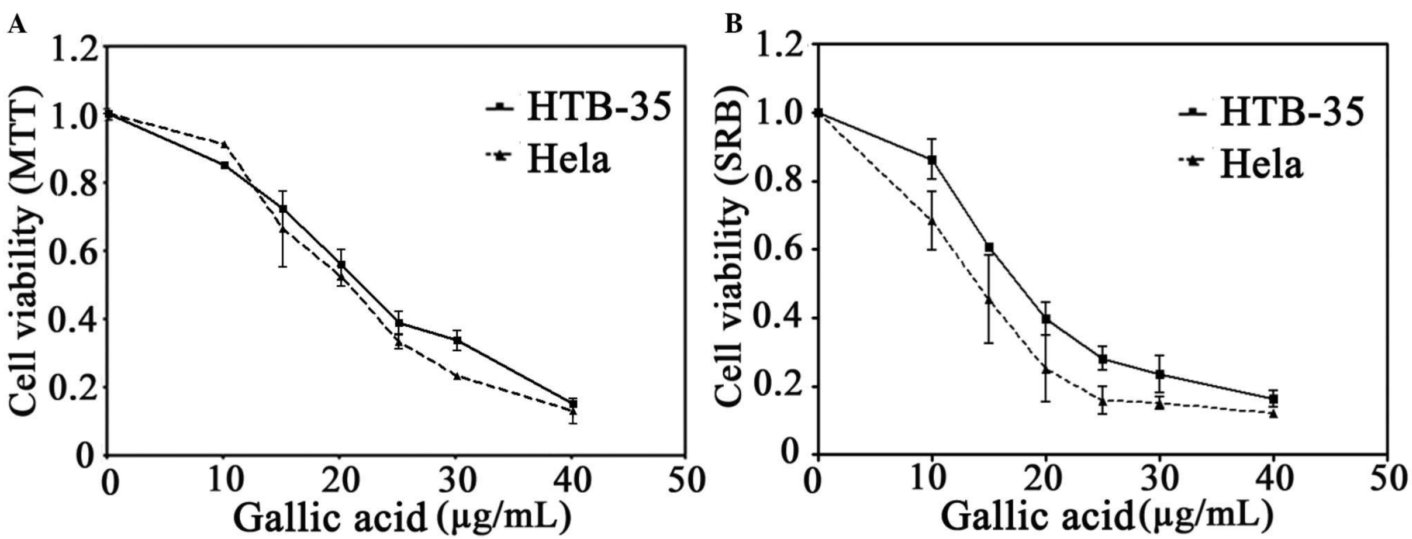


Figure 3: UV/VIS absorbance spectra of MTT salt with and without K:D-Rib solution. black line: MTT salt UV/VIS spectrum; red line: formazan salt after solubilization (ROCHE data sheet).

https://openi.nlm.nih.gov/detailedresult.php?img=PMC3173286_1475-2867-11-30-3&req=4



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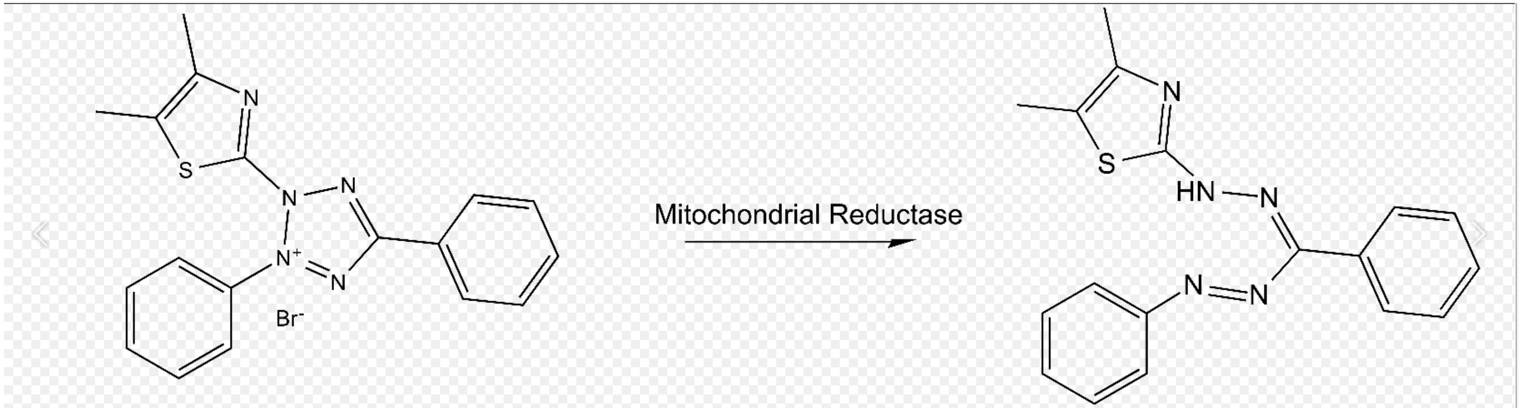
Cell Viability & Proliferation Assays

MTT assay



3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
(MTT)

(*E*,*Z*)-5-(4,5-dimethylthiazol-2-yl)-1,3-diphenylformazan
(Formazan)

A variety of tetrazolium compounds have been used to detect viable cells. The most commonly used compounds include: **MTT, MTS, XTT, and WST-1**.

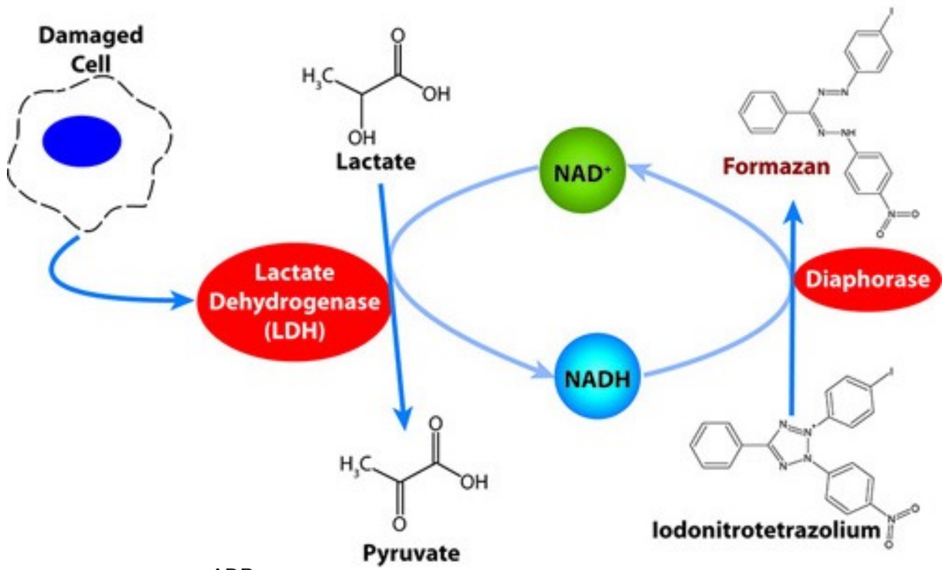
These compounds fall into two basic categories:

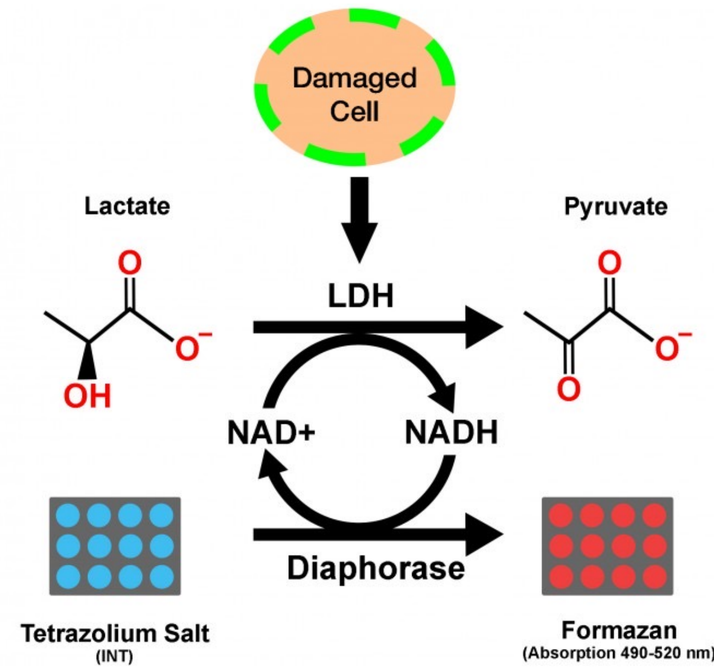
1) MTT which is positively charged and readily penetrates viable eukaryotic cells and
2) those such as **MTS, XTT, and WST-1** which are negatively charged and do not readily penetrate cells. The latter class (MTS, XTT, WST-1) are typically used with an intermediate electron acceptor that can transfer electrons from the cytoplasm or plasma membrane to facilitate the reduction of the tetrazolium into the colored formazan product.

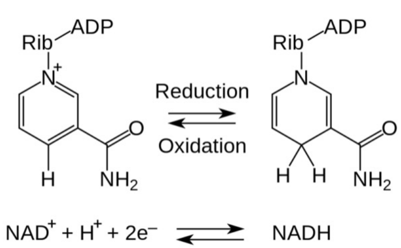
Death assay

LDH Cytotoxicity Detection Kit- Easy, Colorimetric Test for Cell Death

The easiest way to measure cell death is by measuring lactate dehydrogenase (LDH), a stable cytoplasmic enzyme which is present in all cells **but only released when the plasma membrane is damaged**. The LDH Cytotoxicity Detection Kit provides a simple and precise colorimetric assay for LDH activity: a two-step enzymatic reaction creates a formazan dye that is easily measured by A492.







NADH（Nicotinamide adenine dinucleotide）是一种化学物质，是烟酰胺腺嘌呤二核苷酸的还原态，还原型辅酶 I 。N指烟酰胺，A指腺嘌呤，D是二核苷酸。

LIVE/DEAD® Cell Viability Assay

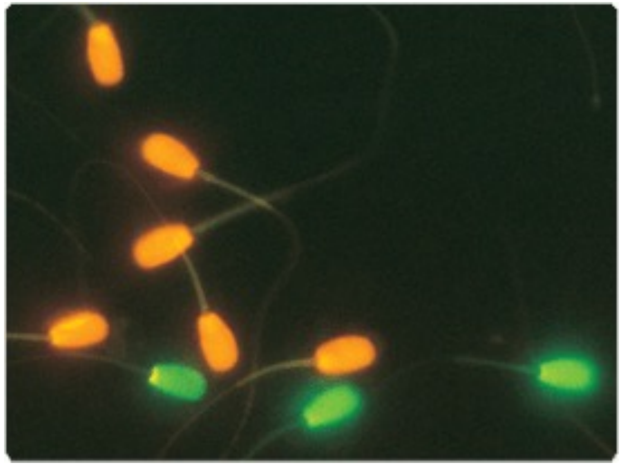
Selection guides for LIVE/DEAD Cell Viability Assay Kits

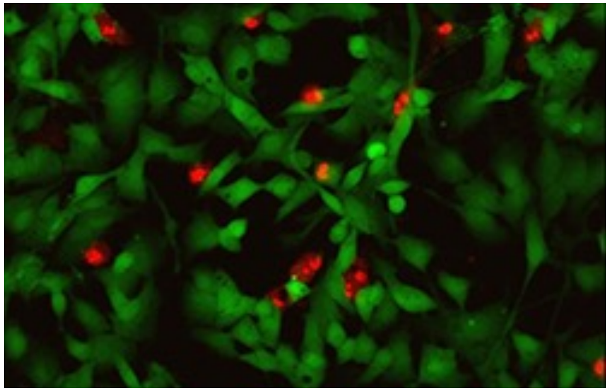
Kits for mammalian cells | Kits for yeast and fungi | Kits for bacteria

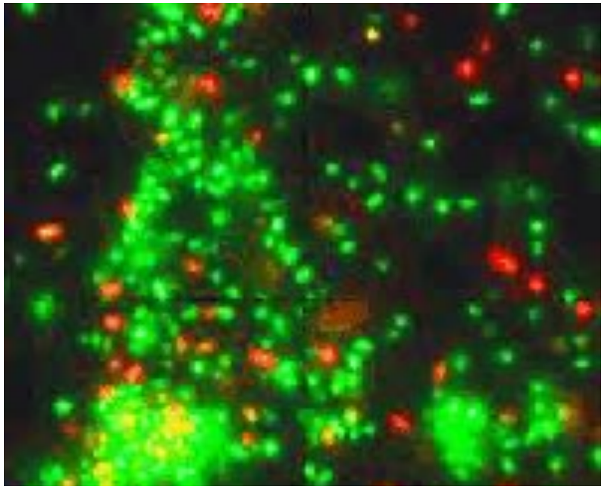
LIVE/DEAD Kits for mammalian cells

Kit Name	Platform*	Fluorescent Dyes	Ex/Em (nm)	Em Colors	Cat. No.
LIVE/DEAD Viability/Cytotoxicity Kit for mammalian cells	FC, FM, M	calcein AM ethidium homodimer-1	494/517 517/617	Green (Live) Red (Dead)	L3224
LIVE/DEAD Cell-Mediated Cytotoxicity Kit for animal cells, 2000 assays	FC, FM, M	$DiOC_{18}(3)$ propidium iodide	484/501 536/617	Green (Live) Red (Dead)	L7010
LIVE/DEAD Sperm Viability Kit 200–1,000 assays	FC, FM	Applied Biosystems SYBR 14 dye propidium iodide	485/517 536/617	Green (Live) Red (Dead)	L7011
LIVE/DEAD Cell Vitality Assay Kit C_{12}-resazurin/SYTOX Green 1,000 assays	FC, FM, M	SYTOX Green dye C12-resazurin	488/530 488/575	Green (Dead) Red (Live)	L34951

*FC = flow cytometry; FM = fluorescence microscopy; M = microplate assay



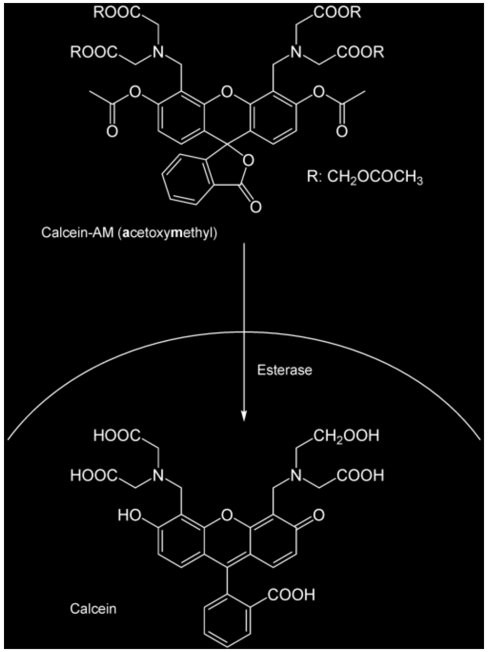




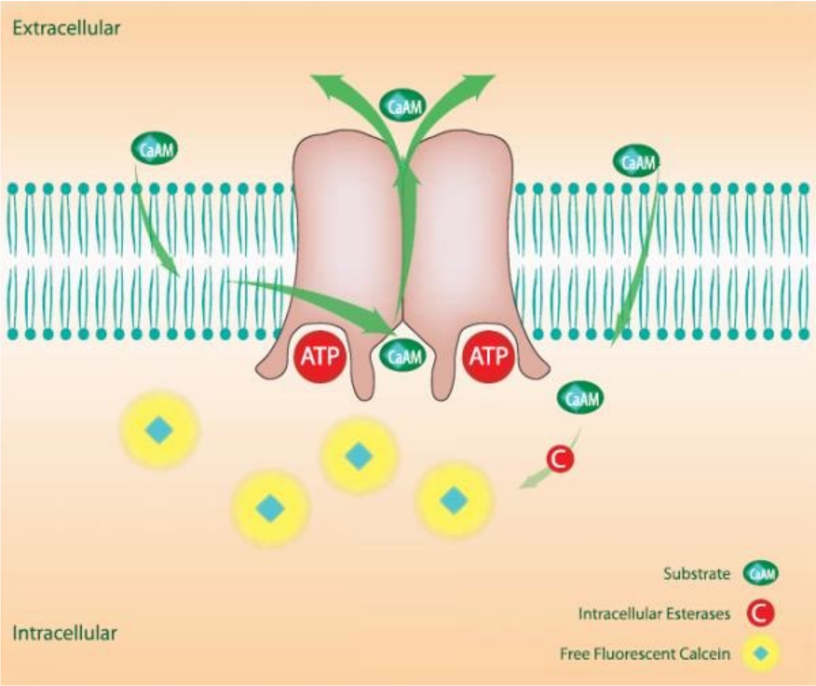
Homework: search and remember the structures for the chemicals in this table

Calcein AM

- Calcein = Free Fluorescent Dye (Green)
  Requires esterase activity
- CaAM actively transported out of cell
  Release during cytolysis



http://upload.wikimedia.org/wikipedia/commons/9/90/Calcein-AM.png



http://www.xenotechllc.com/_FileLibrary/PageFile/204/TR_Calceincell_600x503.jpg

Homework；

1. For in vitro and in vivo tests, we talked about 11 kinds of tests, list the names of four kinds of tests and the function or potential problems of one kind of test.

2. Implanted sites in animal models need test 4 basic types of tissues, list the names.

3. Three morphological cell culture assays are primarily used for evaluating biocompatibility, what are these assays?

4. How many kinds toxicity or cytotoxicity assays we talked, either by morphological changes, absorption, or emission approaches? List the names or if possible remember their mechanisms.

5. Homework: search and remember the structures for the chemicals in the table of live-dead kit (on page 58)